College, Debt, and Gender: The Influence of Outside Options on Higher Education

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Introduction

A seemingly paradoxical pattern has emerged in higher education

- Roughly 40% of women hold a BA by age 25–30, vs. roughly a third of men
- At the same time, women have worse post-college outcomes
  - Lower post-graduate earnings
  - Lower labor force participation
  - Higher student debt balances

Motivating Questions:

- What explains these gender gaps in higher education?
- What are the implications for gender equality and higher education?

This Paper

Main Mechanism: worse outside options raise net return to college for women.

New data on sources of these gender gaps in education

- Excess female matriculation from lower income, worse outside option localities
- Disproportionately sort into lower quality, less expensive schools...
- ...but still end up with more student debt and lower earnings.

New implications for gender equality and higher education

→ Gender earnings gap may move in opposite direction of female welfare.

→ Medium term fluctuations in college premium due to compositional effects, unrelated to labor demand.

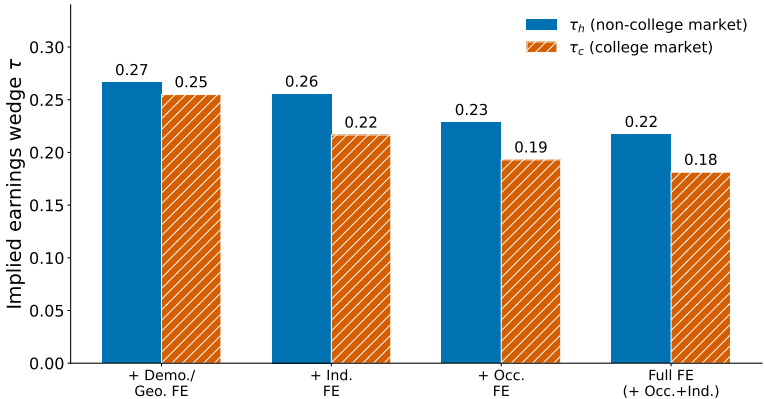
Today's talk

1. Empirical evidence on sources of excess female matriculation
2. Illustrative model
3. Additional facts on college sorting and debt
4. Quantitative model extensions
5. Calibration strategy
6. Preliminary results

Empirical Evidence on Sources of Excess Female Matriculation

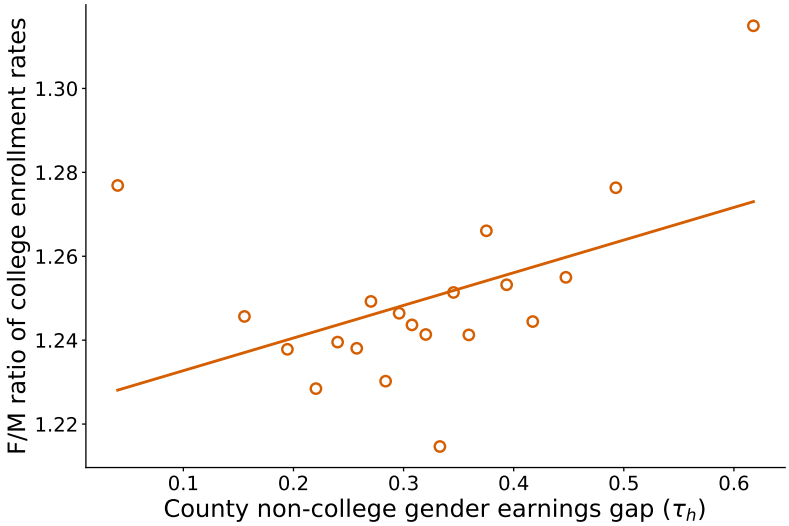
Discrimination in College and Non-College Markets

Hours



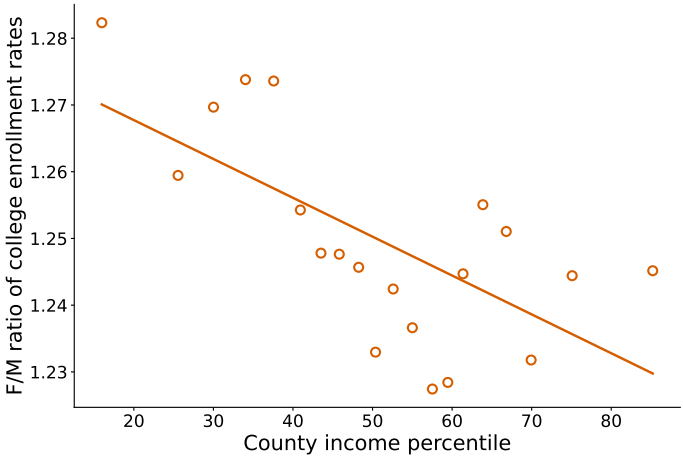
Source: CPS ASEC.

Source of Excess Female Matriculation



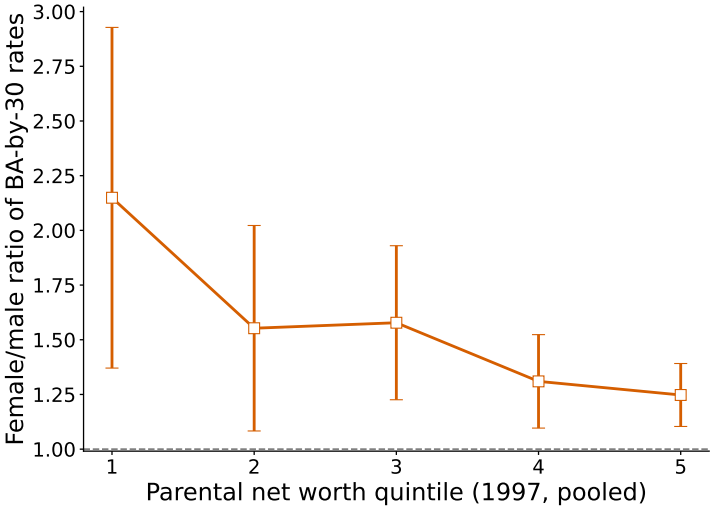
Source: ACS.

Source of Excess Female Matriculation



Source: ACS.

Source of Excess Female Matriculation



Source: NLSY97.

Illustrative Model

An Illustrative Model

High School Graduate.

$$V_g^h = \max\ u(c) \quad \text{s.t.} \quad c \leq w_h(1-\tau_h)$$

College Graduate.

$$V_g^c(\varphi) = \max\ u(c) \quad \text{s.t.} \quad c \leq w_c(1-\tau_c) - \frac{1}{\varphi}p$$

Illustrative Model

Matriculation.

$$\Pi_g \equiv \mathsf{P}\left(\mathsf{V}_g^c(\varphi) > \mathsf{V}_g^h\right)$$

Student Debt.

$$\mathsf{SD}_g \equiv \mathbb{E}\left[\frac{1}{\varphi}\, p \,\middle|\, \text{enrolled in college}\right]$$

Illustrative Model

For simplicity, heterogeneity in financial support $\varphi \sim \text{Pareto}(\chi, \theta)$

Matriculation.

$$\Pi_g = \left(\chi \left[\frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{p}\right]\right)^{\theta}$$

Student Debt.

$$\text{SD}_g = \frac{\theta}{1+\theta}\left[(1-\tau_c)w_c - (1-\tau_h)w_h\right]$$

Illustrative Model

Matriculation Gap.

$$\frac{\Pi_w}{\Pi_m} = \left[\frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{(w_c - w_h)}\right]^{\theta}$$

Student Debt Gap:

$$\frac{SD_w}{SD_m} = \frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{w_c - w_h}$$

Aggregate Implications

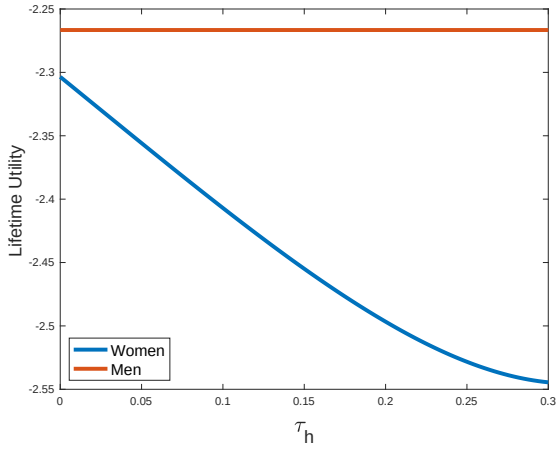
$$\textbf{Agg. Gender Earnings Gap} \quad = \quad \frac{(1-\Pi_w)(1-\tau_h)w_h \;+\; \Pi_w(1-\tau_c)w_c}{(1-\Pi_m)w_h \;+\; \Pi_m w_c}$$

$$\textbf{College Earnings Premium} \quad = \quad \frac{\Pi_w^c(1-\tau_c) \;+\; (1-\Pi_w^c)}{\Pi_w^h(1-\tau_h) \;+\; (1-\Pi_w^h)} \times \frac{w_c}{w_h}$$

where $\quad \Pi_g^c \equiv \dfrac{\Pi_g}{\Pi_w + \Pi_m} \quad , \quad \Pi_g^h \equiv \dfrac{1-\Pi_g}{1-\Pi_w + 1 - \Pi_m}$

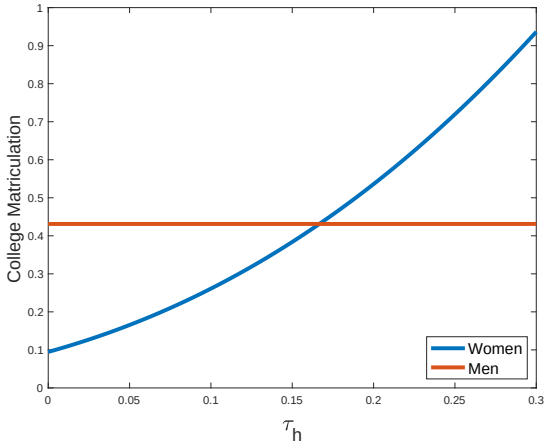
Welfare Gap

$$\text{Welfare} \;=\; (1-\Pi_g)\times V_g^h \;+\; \Pi_g \times \mathbb{E}\left[\, V_g^c(\varphi) \;\middle|\; \text{enrolled in college}\right]$$



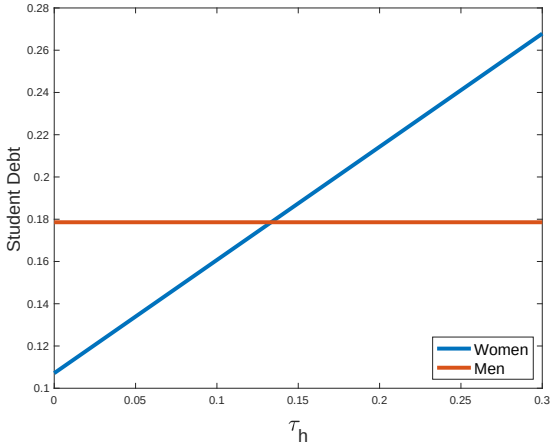
College Matriculation

$$\Pi_g = \left[\frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{p/\chi} \right]^{\theta}$$



Student Debt Gap

$$\mathbb{E}\left[\frac{1}{\varphi}p \,\middle|\, \text{enrolled in college}\right] = \frac{\theta}{1+\theta}\left[(1-\tau_c)w_c - (1-\tau_h)w_h\right]$$



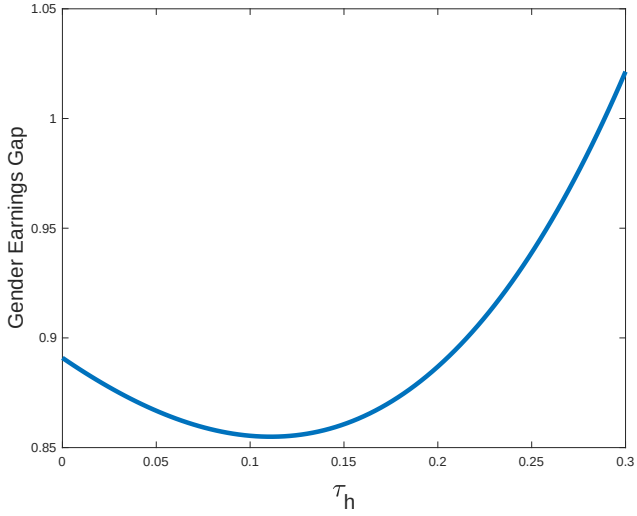
Aggregate Gender Earnings Gap

$$\text{Agg. Gender Earnings Gap} \;=\; \frac{(1-\Pi_w)(1-\tau_h)w_h \;+\; \Pi_w(1-\tau_c)w_c}{(1-\Pi_m)w_h \;+\; \Pi_m w_c}$$

As τ_h increases, the aggregate gender earnings gap

- **Direct:** falls because it reduces earnings of non-college women, $(1-\tau_h)w_h$
- **Compositional:** rises because it increases women's matriculation, Π_w

Aggregate Gender Earnings Gap



College Earnings Premium

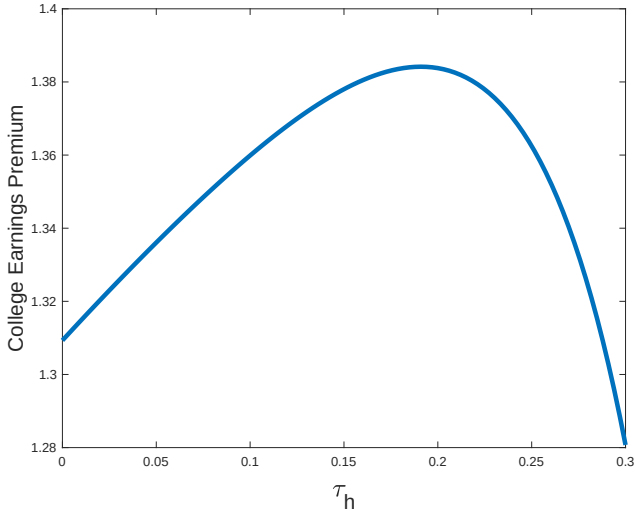
$$\text{College Earnings Premium} \;=\; \frac{\Pi_w^c(1-\tau_c) \;+\; \Pi_m^c}{\Pi_w^h(1-\tau_h) \;+\; \Pi_m^h} \times \frac{w_c}{w_h}$$

$$\text{where} \qquad \Pi_g^c \equiv \frac{\Pi_g}{\Pi_w + \Pi_m} \qquad , \qquad \Pi_g^h \equiv \frac{1-\Pi_g}{1-\Pi_w + 1 - \Pi_m}$$

As τ_h increases, the college earnings premium

- **Direct:** increases by reducing earnings of non-college women, $(1-\tau_h)w_h$
- **Compositional:** decreases because women become greater share of college population Π_w^c and face a gender earnings gap $(1-\tau_c)w_c < w_c$.

Aggregate College Earnings Premium



What about the impact of Student Debt?

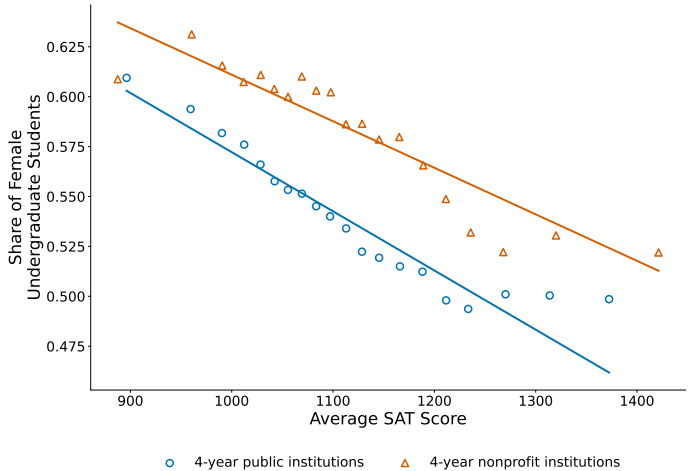
Graduating with more student debt can affect post-college earnings,

- Job Amenities and Initial Earnings (Rothstein Rouse 2011,Mongey Liu 2024)
- Occupation Choice and Returns to Experience (Alon, Cox, and Kim 2024)
- College Major Choice (Hampole 2024)
- Housing and Entrepreneurship (Folch and Mazzone 2022, Morazzoni 2024)
- Job Search and Matching (Herkenhoff, Phillips, Cohen-Cole 2024)

As well as the type of college attended...

Female Sorting in Higher Education

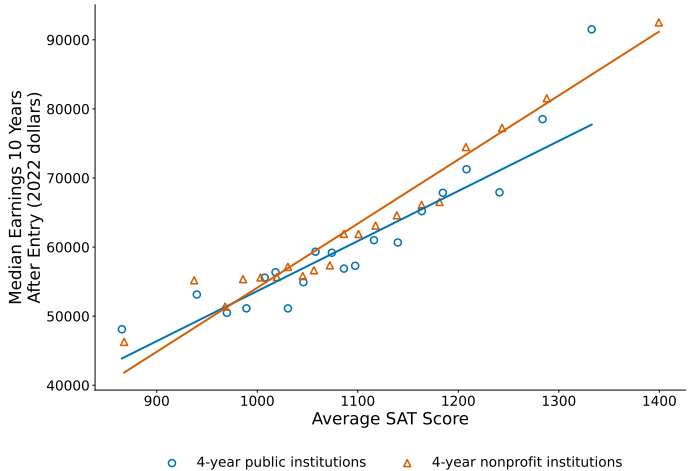
Women attend lower quality schools on average ...



Source: College Scorecard.

Female Sorting in Higher Education

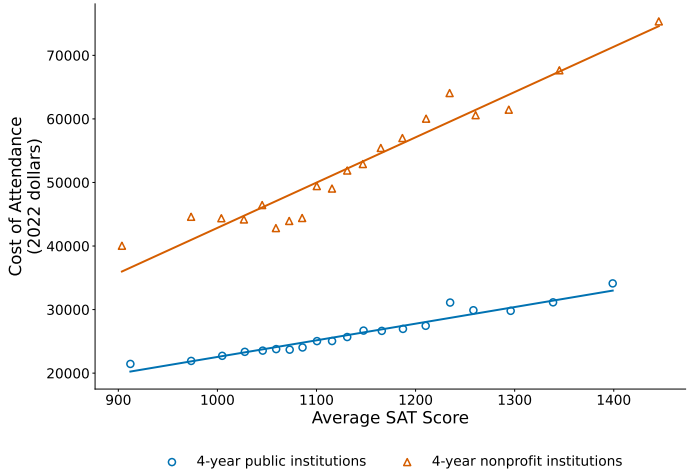
...resulting in them having lower earnings after graduation



Source: College Scorecard.

Female Sorting in Higher Education

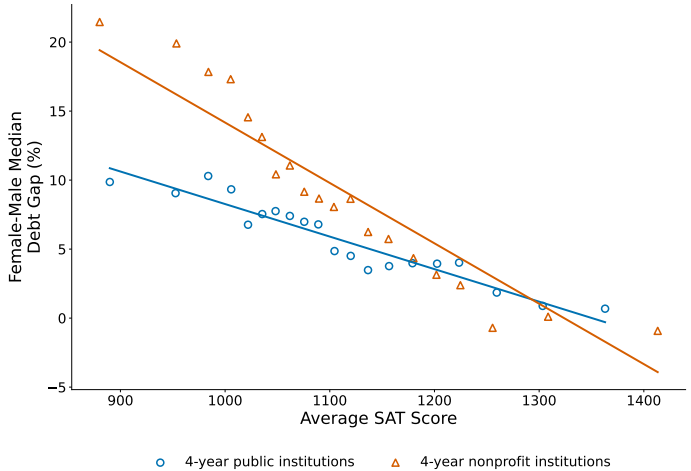
...and while these schools cost less to attend



Source: College Scorecard.

Female Sorting in Higher Education

...women still end up with more student debt relative to male peers



Source: College Scorecard.

Quantitative Model Extensions

Quantitative Model Extensions

- **Initial Household Heterogeneity** in parental wealth (a_0), ability (z), and gender.
- **Endogenous Impact of Student Debt on Earnings** arising from school quality offers, post-grad investment, and financial frictions.
- **Young and Old Labor** with terminal retirement state
- **Production Sector** to endogenize wages w^C, w^H

$$Y = \left[A_C H_C^{\eta} + A_H H_H^{\eta} \right]^{\frac{1}{\eta}}$$

- **Endogenous Labor Supply** with heterogeneity across men and women
- **Residuals:** unobserved costs τ_s and college preferences κ_g

Young Worker

$$V_y^{g,e}(h,a) \;=\; \max_{c,s,n,a'} \;\; u^{g,e}(\,c\,,\,n\,,\,s\,) \;+\; \beta V_o^{g,e}(h',a')$$

subject to

$$\begin{aligned} c \;+\; a' \;&\leq\; (1-\tau^{g,e})\, w^e (1-s) hn \;+\; (1+r)a \\ h' \;&=\; B^e (hs)^{\alpha} + (1-\delta)h \\ a' \;&\geq\; 0 \end{aligned}$$

with Greenwood, Hurcowitz, Huffman (1988) preferences

$$u^{g,e}(\,c\,,\,n\,,\,s\,) \;=\; \frac{1}{1-\rho}\left(c \;-\; \psi^{g,e}(1-s)\,\frac{n^{1+\nu}}{1+\nu} \right)^{1-\rho}$$

Student debt enters as negative initial assets; $a' \geq 0$ forces repayment.

Old Worker

$$V_o^{g,e}(h,a) \;=\; \max_{c,n,a'} \; u^{g,e}(\,c\,,\,n\,,\,0) \;+\; \beta V_R(a')$$

subject to

$$\begin{aligned} c + a' \;&\leq\; (1-\tau^{g,e})\, w^e h n + (1+r)a \\ a' \;&\geq\; 0 \end{aligned}$$

with (reduced-form) terminal retirement state

$$V_R(a) \;=\; \mathcal{R}\,\frac{(a+\bar{a})^{1-\rho}}{1-\rho}$$

Higher Education: School Offers and Costs

Initial Heterogeneity: ability and parental wealth, common to men and women

$$(\log z,\ \log a_0)\ \sim\ \mathcal{N}(\mu_F,\ \Sigma_F)$$

School Offer: each student receives one offer q and accepts or rejects it

$$\log q \mid z, a_0\ \sim\ \mathcal{N}\Big(\mu_o + \beta_z \log z + \beta_a \log(1 + a_0),\ \sigma_o^2\Big)$$

Cost and Borrowing:

$$a_{\text{entry}}\ =\ a_0 - (1 + \tau_s^g)\left(p + \omega q^{\zeta}\right), \qquad -\min\{0,\ a_{\text{entry}}\}\ \leq\ d_{\max}$$

Higher Education: College Choice

Extensive Margin: accept offer q if

$$V_C^g(z, a_0, q) \;+\; \kappa^g(a_0) \;+\; \epsilon_C \;\geq\; V_H^g(z, a_0) \;+\; \epsilon_H$$

where

$$V_C^g(z, a_0, q) \;=\; \mathbb{E}_\varepsilon\Big[V_y^{g,C}\big(h^C,\, a_{\text{entry}}\big)\Big], \qquad h^C \;=\; z^\xi(1 + q^\gamma)\,\varepsilon$$

$$V_H^g(z, a_0) \;=\; V_y^{g,H}\big(h^H,\, a_0\big), \qquad h^H \;=\; z^\xi$$

- ϵ_C, ϵ_H: Gumbel taste shocks with scale σ_κ
- $\kappa^g(a_0)$: college taste by gender and parental wealth quintile
- Unaffordable offer is a rejection, not a trade-down

Production Technology

Production uses skilled and unskilled labor

$$Y = \left[A_C H_C^{\eta} + A_H H_H^{\eta} \right]^{\frac{1}{\eta}}, \qquad w^e = Y^{1-\eta} A_e H_e^{\eta - 1}$$

where H_e is total effective hours of education group e,

$$H_e = \mathbb{E}\left[(1-s) hn \,|\, e \right]$$

which implicitly average over age groups and gender.

Wages are fixed in the calibration and clear the labor market in the counterfactuals.

Aggregate Implications

Aggregates now depends on changes in matriculation and earnings

$$\textbf{Agg. Gender Earnings Gap} \quad = \quad \frac{(1-\Pi_w)\,Y_w^h \;+\; \Pi_w\,Y_w^c}{(1-\Pi_m)\,Y_m^h \;+\; \Pi_m\,Y_m^c}$$

$$\textbf{College Earnings Premium} \quad = \quad \frac{\Pi_w^c\,Y_w^c \;+\; (1-\Pi_w^c)\,Y_m^c}{\Pi_w^h\,Y_w^h \;+\; (1-\Pi_w^h)\,Y_m^h}$$

where earnings vary endogenously with education, gender, student debt and age

$$Y^e = \mathbb{E}\left[(1-\tau_e)w_e(1-s^*)h^*n^* \,\middle|\, \text{education level } e^*\right]$$

Counterfactual Exercises

Objective: Quantify the impact of τ on the (i) education gaps, (ii) college earnings premium, and (iii) aggregate gender earnings gap.

Strategy: Structural estimation of τ_h,τ_c using stylized facts and causal evidence.

Today: Steady state counterfactuals (general equilibrium)

- Set $\tau_h = 0$
- Set $\tau_c = 0$
- Set $\tau_h = \tau_c = 0$

Work in Progress: Transition dynamics, 1980–2025.

Calibration Strategy

Calibration Strategy: Steady State Targets

Match Key Moments by Education and Gender:

- Earnings (w^e, $\tau^{f,e}$) and lifecycle earnings growth (B^e)
- Labor Supply ($\psi^{g,e}$)
- Matriculation, by gender and parental wealth (κ^g)
- Student Debt by gender (p, ω, τ_s)

Match Heterogeneity in College Earnings:

- Earnings gradient of college quality (ξ, γ)
- Dispersion of young college earnings (σ^2)

Match Moments of Asset Distribution:

- Wealth at labor market entry ($\mu_{F,a}$, $\Sigma_{F,aa}$)
- Assets at retirement ($\mathcal{R}$)

Calibration Strategy: Indirect Inference on Key Elasticities

How do changes in outside options effect college matriculation?

- Evidence from housing, fracking, and China shocks Evidence
- Pins down distribution of education taste shocks σ_κ

Target: enrollment response to non-college expected income

- Fracking IV (preferred estimate): -0.190

Calibration Results: Key Targets

Full Table | External Parameters

Parameter	Moment	Data	Model	Diff. (%)
$\sigma_\kappa = 0.043$	Attendance response to expected income	-0.190	-0.187	$+1.4$
$\kappa_0^m = -0.114$	BA attainment, men	0.3246	0.3246	0.0
$\kappa_0^f = -0.088$	BA attainment, women	0.3961	0.3961	0.0
$w^C = 104.4$	Mean earnings, college men	217.1	217.4	$+0.1$
$w^H = 112.8$	Mean earnings, HS men	105.3	103.8	-1.4
$\tau^{f,C} = 0.198$	Gender earnings ratio, college	0.7583	0.7580	0.0
$\tau^{f,H} = 0.226$	Gender earnings ratio, HS	0.7442	0.7365	-1.0
$p = 0.241$, $\omega = 2.582$	Mean debt, men	0.937	0.939	$+0.2$
$\tau_s = 0.083$	Debt gap, F $-$ M	0.2219	0.2224	$+0.2$

Counterfactuals

Counterfactuals: Steady State

Other Gaps

Moment	Data	Model	$\tau_h = 0$	$\tau_c = 0$	$\tau_h = \tau_c = 0$
Matriculation Gap	1.220	1.220	1.010	1.350	1.124
Student Debt Gap	1.237	1.237	1.144	1.288	1.195
Gender Earnings Gap	0.795	0.792	0.905	0.980	1.106
College Earnings Premium	2.033	2.073	2.081	2.150	2.152

Counterfactuals: Steady State

Moment	Data	Model	$\tau_h = 0$	$\tau_c = 0$	$\tau_h = \tau_c = 0$
Matriculation Gap	1.220	1.220	1.010	1.350	1.124
Student Debt Gap	1.237	1.237	1.144	1.288	1.195
Gender Earnings Gap	0.795	0.792	0.905	0.980	1.106
College Earnings Premium	2.033	2.073	2.081	2.150	2.152

$\rightarrow \tau_h = 0$ accounts for $(22.0 - 1.0)/22.0 \approx 95\%$ of gender matriculation gap

Counterfactuals: Steady State

Moment	Data	Model	$\tau_h = 0$	$\tau_c = 0$	$\tau_h = \tau_c = 0$
Matriculation Gap	1.220	1.220	1.010	1.350	1.124
Student Debt Gap	1.237	1.237	1.144	1.288	1.195
Gender Earnings Gap	0.795	0.792	0.905	0.980	1.106
College Earnings Premium	2.033	2.073	2.081	2.150	2.152

$\rightarrow \tau_h = 0$ accounts for $(23.7 - 14.4)/23.7 \approx 39\%$ of gender student debt gap

Counterfactuals: Steady State

Moment	Data	Model	$\tau_h = 0$	$\tau_c = 0$	$\tau_h = \tau_c = 0$
Matriculation Gap	1.220	1.220	1.010	1.350	1.124
Student Debt Gap	1.237	1.237	1.144	1.288	1.195
Gender Earnings Gap	0.795	0.792	0.905	0.980	1.106
College Earnings Premium	2.033	2.073	2.081	2.150	2.152

$\rightarrow \tau_h = 0$ accounts for $(1 - 0.792) - (1 - 0.905) \approx 11$ pp ($\approx 54\%$) of gender earnings gap

Counterfactuals: Steady State

Moment	Data	Model	$\tau_h = 0$	$\tau_c = 0$	$\tau_h = \tau_c = 0$
Matriculation Gap	1.220	1.220	1.010	1.350	1.124
Student Debt Gap	1.237	1.237	1.144	1.288	1.195
Gender Earnings Gap	0.795	0.792	0.905	0.980	1.106
College Earnings Premium	2.033	2.073	2.081	2.150	2.152

$\rightarrow \tau_h = 0$ leaves college earnings premium essentially unchanged once wages adjust

Additional Exercises

I. Decomposing the Mechanisms: *exogenous* changes in τ versus *endogenous* response of matriculation, wages, human capital.

II. Transition Dynamics, 1980-2025

Conclusion

- Analyze the emergence of these seemingly paradoxical gender gaps
- Document significant role of relative differences in outside options
- Quantify implications for the composition and earnings of higher education
- Quantify implications for measurement and dynamics of gender inequality

Work in Progress:

- Transition Path dynamics for gender gap, college premium, and matriculation.
- Micro evidence on sources of student debt gaps and financial consequences (e.g. other forms of credit, bankruptcy, delinquency, repayment, etc)

Appendix

Externally Calibrated Parameters

Back

Parameter	Symbol	Value	Source
Inverse IES	ρ	2	Standard
Inverse Frisch elasticity	ν	2	Standard
Discount factor (20 years)	β	$0.99^{20} = 0.818$	1% annual
Interest rate (20 years)	r	$1.03^{20} - 1 = 0.806$	3% annual
Ben-Porath curvature	α	0.60	Literature; below 2/3
Depreciation (20 years)	δ	0.367	Huggett, Ventura, Yaron (2011), 2.26%/yr
Cost curvature	ζ	1.82	Scorecard cost-quality elasticity
Federal loan limit	$d_{\max}$	3.1	Statute ($31,000)
Offer gradient in ability	β_z	0.4288	Scorecard, Bayes inversion
Offer dispersion	σ_o	0.0805	Scorecard, Bayes inversion
Offer gradient in wealth	β_a	0	Not yet estimated
Offer intercept	μ_o	-0.125	Normalization
Disutility, HS men	$\psi^{m,H}$	106.7	Hours normalization
Variance of log ability	$\Sigma_{F,zz}$	0.0461	SAT 2019
Ability-wealth correlation	$\mathrm{corr}(\log z, \log a_0)$	0.3648	NLSY97
CES curvature	η	0.333	Jones (2014); counterfactuals only

Internally Calibrated Parameters and Model Fit (1/3)

Back

Parameter	Moment	Data	Model	Diff. (%)
Earnings and hours (CPS ASEC 2010–2022)				
$w^C = 104.4$	Mean earnings, college men	217.1	217.4	+0.1
$w^H = 112.8$	Mean earnings, HS men	105.3	103.8	−1.4
$\tau^{f,C} = 0.198$	Gender earnings ratio, college	0.7583	0.7580	0.0
$\tau^{f,H} = 0.226$	Gender earnings ratio, HS	0.7442	0.7365	−1.0
$B^C = 1.410$	Lifecycle earnings growth, college	1.265	1.265	0.0
$B^H = 0.989$	Lifecycle earnings growth, HS	1.147	1.141	−0.5
$\sigma^2 = 0.099$	Var log earnings, young college	0.3948	0.3943	−0.1
$\psi^{m,C} = 195.1$	Hours, college men (rel. HS men)	1.034	1.032	−0.2
$\psi^{f,C} = 170.3$	Hours, college women (rel. HS men)	0.984	0.9843	0.0
$\psi^{f,H} = 91.9$	Hours, HS women (rel. HS men)	0.9472	0.9493	+0.2
School quality (College Scorecard)				
$\xi = 0.659$, $\gamma = 0.050$	Male quality-earnings slope	1.584	1.585	+0.1
Student debt (NPSAS 2012 and 2016)				
$p = 0.241$, $\omega = 2.582$	Mean debt, men	0.937	0.939	+0.2
$\tau_s = 0.0827$	Debt gap, F − M	0.2219	0.2224	+0.2

Internally Calibrated Parameters and Model Fit (2/3)

Back

Parameter	Moment	Data	Model	Diff. (%)
College attendance (CPS; fracking IV)				
$\kappa_0^m = -0.1141$	BA attainment, men	0.3246	0.3246	0.0
$\kappa_0^f = -0.0880$	BA attainment, women	0.3961	0.3961	0.0
$\sigma_\kappa = 0.043$	Attendance response to expected income	−0.190	−0.187	+1.4
Attendance by parental-wealth quintile (NLSY97, BA by 30)				
κ^m profile	Men, Q1	0.0922	0.0922	0.0
$\kappa_2^m = 0.028$	Men, Q2	0.1579	0.1591	+0.8
$\kappa_3^m = 0.055$	Men, Q3	0.2575	0.2611	+1.4
$\kappa_4^m = 0.093$	Men, Q4	0.4460	0.4536	+1.7
$\kappa_5^m = 0.137$	Men, Q5	0.6693	0.6869	+2.6
κ^f profile	Women, Q1	0.1730	0.1729	0.0
$\kappa_2^f = 0.014$	Women, Q2	0.2140	0.2151	+0.5
$\kappa_3^f = 0.044$	Women, Q3	0.3547	0.3573	+0.7
$\kappa_4^f = 0.072$	Women, Q4	0.5099	0.5151	+1.0
$\kappa_5^f = 0.119$	Women, Q5	0.7289	0.7456	+2.3

Internally Calibrated Parameters and Model Fit (3/3)

Back

Parameter	Moment	Data	Model	Diff. (%)
Wealth and ability (SCF+; SAT 2019; NLSY97)				
$\mu_{F,a} = -0.8322$	Mean wealth at entry (25–30)	1.710	1.786	+4.4
$\Sigma_{F,aa} = 3.6499$	Sd of wealth at entry (25–30)	15.20	14.60	−4.0
$\mathcal{R} = 0.380$	Mean assets (65–70)	42.94	43.22	+0.6
$\mu_{F,z} = -0.00268$	Mean ability	1.028	1.021	−0.7
$\Sigma_{F,zz} = 0.0461$ (imp.)	Var log ability	0.0461	0.0461	0.0
$\Sigma_{F,za} = 0.1496$ (imp.)	Corr(log ability, log wealth)	0.3648	0.3648	0.0

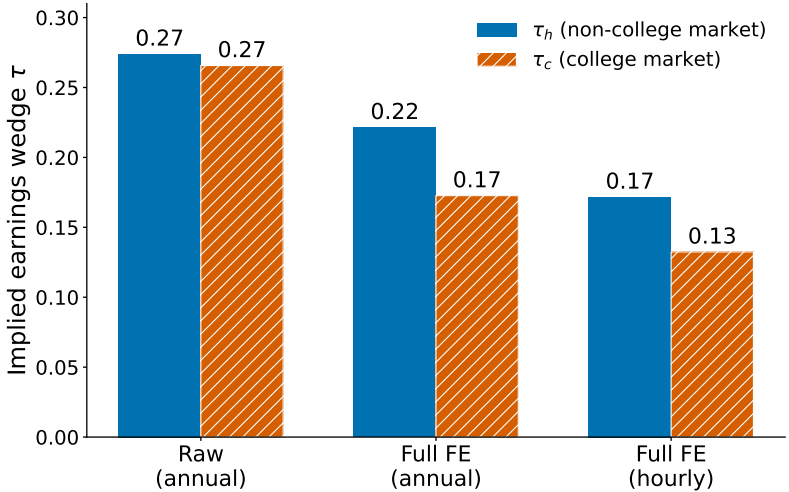
Counterfactuals: One Gendered Block at a Time

Back

Moment	Data	Model	$\tau_h = \tau_c = \bar{\tau}$	ψ averaged	κ averaged	$\tau_s = 0$
Matriculation Gap	1.220	1.220	1.193	1.240	1.059	1.236
Student Debt Gap	1.237	1.237	1.226	1.245	1.137	1.125
Gender Earnings Gap	0.795	0.792	0.781	0.741	0.763	0.795
College Earnings Premium	2.033	2.073	2.073	2.061	2.097	2.062

Discrimination in College and Non-College Markets: Hours

Back



Source: CPS ASEC.

Empirical Evidence on Outside Options and Matriculation

Back

	(1) Housing boom (Charles et al. 2018)	(2) Fracking boom (Cascio-Narayan 2020)	(3) China shock (Autor-Dorn-Hanson 2013)
θ: 2SLS coefficient	-0.129^{***}	-0.190^{**}	-0.171^{***}
	(0.044)	(0.084)	(0.046)
Outside-option shock	Positive	Positive	Negative
Level of variation	Metro area	Commuting zone	Commuting zone
First-stage F	14.9	21.8	28.9
Observations	275	3,110	1,444

A one-percent rise in the non-college outside option lowers 18–24 enrollment by about 0.19pp (fracking, preferred estimate), an elasticity of roughly -0.50. The three designs, despite different shocks/geographies/eras, land within -0.129 to -0.190.

Empirical Strategy: The Housing Boom (Charles et al. 2018)

Back

Identifying variation

- Mid-2000s housing boom raised demand for young non-college labor (construction), most where housing supply was most constrained
- Instrument: size of a *structural break* in each metro's 2000–2005 house-price trend
- Fundamentals move prices smoothly, so a sudden break reflects speculative, housing-specific forces rather than slower local factors that could also affect college-going

Specification and sample

- Long difference: 2000 Census to pooled 2005–2007 ACS, metropolitan-area level
- Outcome: change in the any-college *attainment* share of 18– to 25-year-olds (a stock, not a flow)
- Controls: % college-educated, female labor share, foreign-born share, log population (2000); population-weighted, clustered by state

Result

- $\theta = -0.129^{***}$, first-stage $F \approx 15$
- Reproduces Charles et al.'s published estimate exactly before re-scaling to the outside option

A long difference over the full boom captures attainment (a stock), not enrollment (a flow); since attainment moves less than enrollment to any given shock, this estimate is read as a lower bound relative to the fracking and China-shock designs.

Empirical Strategy: The Fracking Boom (Cascio-Narayan 2020)

Back

Identifying variation

- Post-2007 shale boom raised demand for non-college labor in commuting zones (CZs) sitting on more shale
- Shale reserves are fixed by geology; the drilling technology that made them viable arrived suddenly and everywhere at once
- Instrument: standardized per-capita shale reserves $\times$ Post-2007

Specification and sample

- CZ-by-year panel, 2003–2015 (excludes 2005); outcome: share of 18– to 24-year-olds enrolled in college (a flow)
- Zone and state-by-year fixed effects absorb fixed geology and any statewide shock
- 311 CZs in 17 shale states; weighted by year-2000 population aged 18–24; clustered by CZ

Result (preferred estimate)

- $\theta = -0.190^{**}$, an elasticity of roughly -0.50 at baseline enrollment
- First-stage $F \approx 22$

Event study: exposed and unexposed zones move together before 2008; the first stage then climbs steadily (to about $+13$ log points of expected income per SD of exposure by 2014), and only after does enrollment turn down (about -2pp by 2015).

Empirical Strategy: The China Shock (Autor-Dorn-Hanson 2013)

Back

Identifying variation

- Rising Chinese import competition, 1990–2007, lowered non-college earnings and employment in more import-exposed commuting zones – the only *negative* shock to the outside option among the three designs
- Instrument: growth of Chinese exports to eight other high-income countries, isolating the supply-driven component of China's export surge

Specification and sample

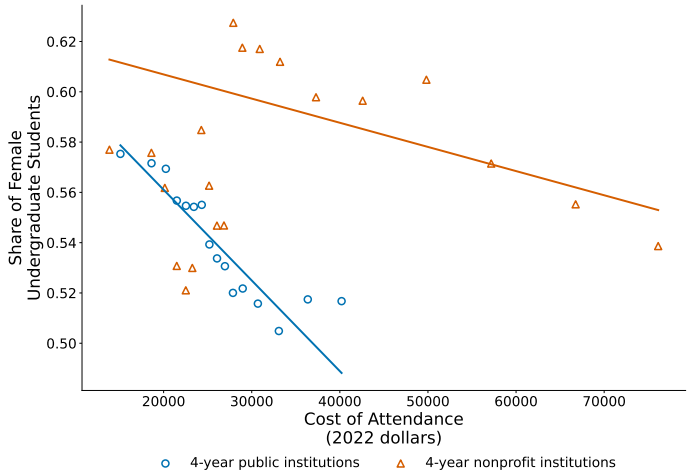
- Stacked panel of two long differences (1990–2000 and 2000–2006/07) with a window fixed effect; outcome: change in the 18–24 enrollment share (a flow)
- Standard ADH controls: manufacturing share, college share, foreign-born share, female employment share, routine-occupation share, offshorability index, Census division
- 722 CZs, 2 stacked windows ($N = 1{,}444$); weighted by ADH weights, clustered by state

Result

- $\theta = -0.171^{***}$, an elasticity of roughly -0.53 at baseline enrollment
- Only design with a usable female-specific first stage: $\theta_{\text{female}} = -0.206^{***}$ vs. an insignificant -0.094 for men

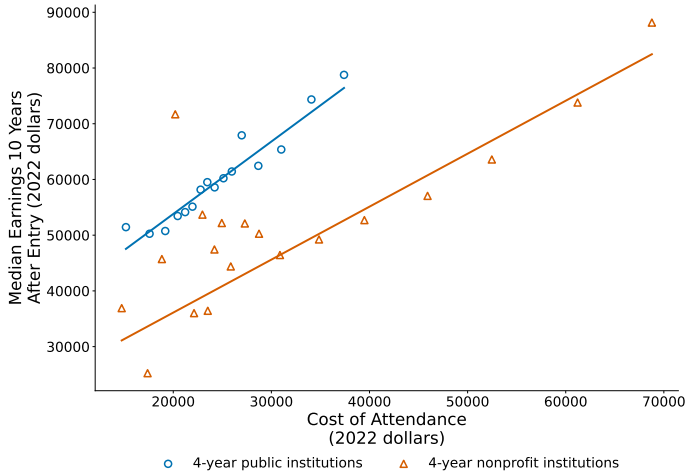
Identification comes entirely from pooling both windows – each alone has too weak a first stage on its own. A timing placebo finds no pre-trend in enrollment.

Female Sorting in Higher Education



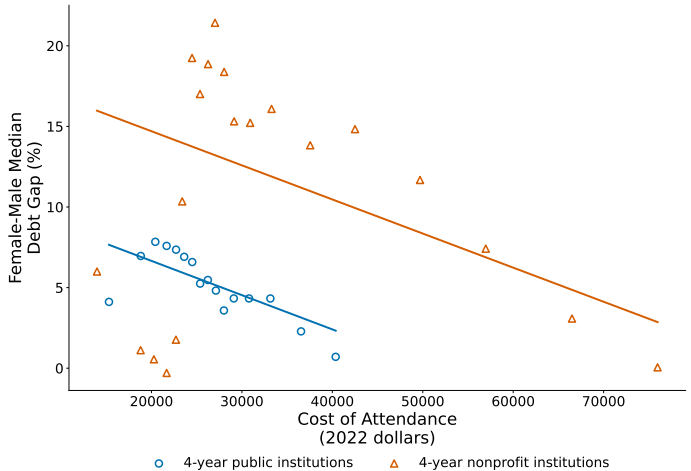
Source: College Scorecard.

Female Sorting in Higher Education



Source: College Scorecard.

Female Sorting in Higher Education

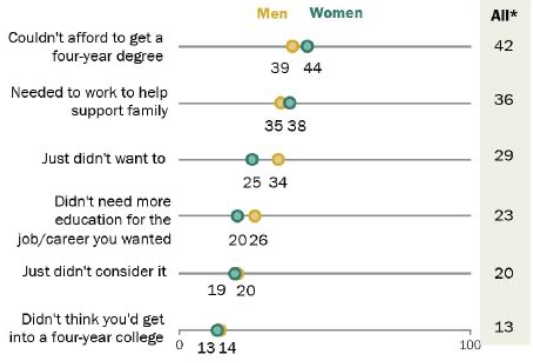


Source: College Scorecard.

Motivating Survey Evidence

About a third of men who haven't completed four years of college say they 'just didn't want to' get a degree

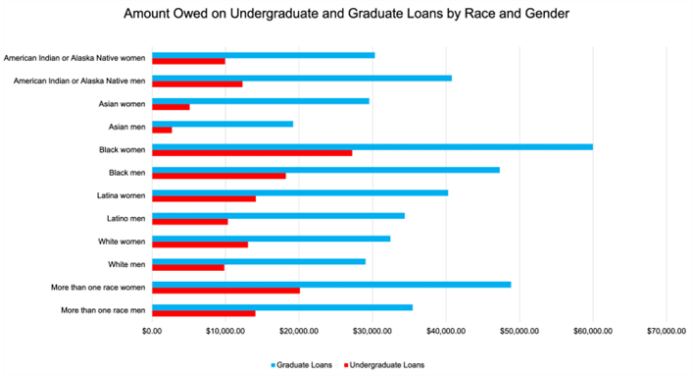
*Among adults who do not have a bachelor's degree and are not enrolled in school, % saying each is a **major** reason why they did not receive a four-year degree*



*All adults who don't have a bachelor's degree and are not enrolled in school.
Source: Survey of U.S adults conducted Oct. 18-24, 2021.

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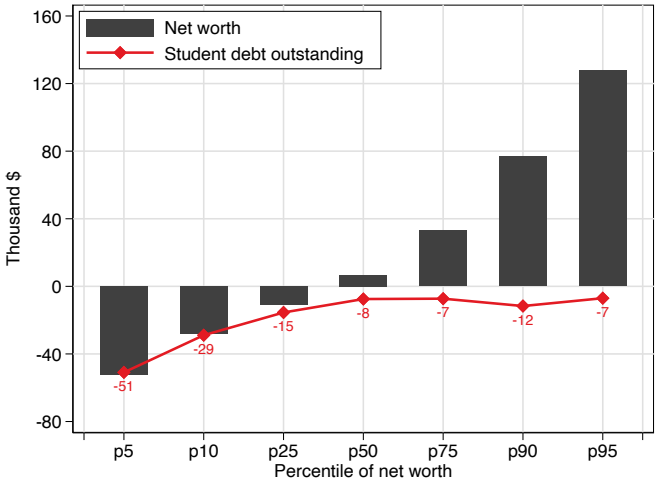
A More General Agenda



A More General Agenda



Student Debt and Net Assets



Source: Survey of Consumer Finances (SCF).

The Data: Linked FAFSA and Credit Records

Back

Administrative Data from California (2006–2023)

- **FAFSA Records:** Universe of federal aid applicants (2006–2015)
  - Detailed parent and student information: income, dependency, school lists
  - Enables parent-child link via SSNs (for dependent students)
- **Credit Bureau Records:** Quarterly tradeline data (2004–2023)
  - Credit cards, student loans, auto loans, mortgages, HELOCs
  - Credit scores, delinquencies, and bankruptcies
  - Parents' and students' financial trajectories tracked before and after college
- **Key Advantage:** Large scale (2 million+) linking parents and kids
- **Sample:** All FAFSA filers in California; 73% attend college

Supplementary Data Sources

In addition to the administrative FAFSA–credit match, the paper uses:

- **IPEDS (Integrated Postsecondary Education Data System)**
  - Provides institutional characteristics: tuition, enrollment, sector, selectivity
- **College Scorecard (U.S. Department of Education)**
  - Includes post-graduation outcomes: earnings, loan repayment, completion rates
- **Decennial Census**
  - Used for geographic and demographic benchmarking of families and neighborhoods
- **American Community Survey (ACS)**
  - Provides supplemental data on income, housing, and education at ZIP and county level